Fire and Smoke Detection Using Computer Vision

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May 4, 2026

***Abstract*—Fire and smoke represent critical hazards with potentially catastrophic consequences for human life, infrastructure, and the natural environment. Traditional sensor-based detection systems suffer from delayed response times and high false-alarm rates, particularly in visually complex scenes [6], [7]. This paper presents a computer-vision-based fire and smoke detection framework built upon the YOLOv8s object detection architecture [5], trained on a large-scale, multi-source dataset comprising 45,252 annotated images drawn from four publicly available repositories: D-Fire, indoor fire scenes, Flame-BD, and Iron Wolf. The merged dataset contains 42,138 fire annotations and 20,757 smoke annotations, yielding a fire-to-smoke ratio of approximately 2.03:1. The dataset was partitioned in a 70/20/10 train/validation/test split, and the model was trained for 45 epochs. The proposed system achieves a mean Average Precision (mAP) of approximately 81% on the held-out test set, demonstrating competitive detection accuracy across diverse indoor and outdoor fire scenarios. Extensive experiments validate that multi-source data merging with careful class remapping substantially improves generalisation over single-source baselines. The results underscore the practical utility of YOLOv8s-based detectors for integration into automated fire-alarm and surveillance systems.**

***Index Terms*—Fire detection, smoke detection, YOLOv8s, object detection, deep learning, computer vision, dataset merging, convolutional neural networks, transfer learning.**

I. Introduction

Fire disasters rank among the most destructive events affecting human societies, causing over 180,000 fatalities annually from burns and smoke inhalation alone [6]. Beyond loss of life, structural and industrial fires inflict billions of dollars in property damage each year and pose severe threats to ecosystems and public infrastructure [6], [7]. Early and reliable detection is therefore the first line of defence against such catastrophes. Traditional detection modalities—ionisation smoke detectors, heat sensors, and sprinkler systems—operate on physical or chemical principles and often react only after combustion has become widespread. These systems are further hampered by high false-alarm rates caused by benign stimuli such as dust, cooking vapour, and steam [6]. Camera-based systems offer a complementary approach: they can cover large areas passively, integrate with existing surveillance infrastructure, and provide actionable visual context to emergency responders in a way that point sensors fundamentally cannot [7].

The emergence of deep convolutional neural networks (CNNs) and, more specifically, real-time single-stage object detectors from the YOLO family [1], [2] has catalysed a new generation of vision-based fire and smoke detectors. These models frame hazard detection as a joint problem—localisation (where is the hazard?) and classification (what class is it?)—producing bounding-box predictions that are directly actionable by suppression or evacuation systems. Recent surveys document rapid progress across both architectures and datasets [6], yet several challenges remain open. Fire and smoke exhibit highly variable appearances: colour, texture, transparency, and scale change dramatically with fuel type, distance, and ambient illumination. Smoke in particular is semi-transparent, amorphous, and can resemble clouds or steam, making it extremely hard for purely local convolutional feature extractors to distinguish from background clutter [2], [6]. Furthermore, publicly available labelled datasets are relatively small and domain-specific, hampering generalisation across the full diversity of real-world fire scenarios [6].

This paper addresses these challenges through three primary contributions:

1) A large-scale merged dataset of 45,252 images assembled from four independent sources with unified class remapping, collision-avoidance file naming, and a rigorous 70/20/10 train/validation/test split.
2) A YOLOv8s detector [5] trained end-to-end on this dataset with COCO-pretrained initialisation and standard mosaic augmentation, achieving $\sim$81% mAP50 on the test partition.
3) A systematic analysis of dataset characteristics—annotation density, class imbalance, and split statistics—and their effect on per-class detection performance.

The remainder of this paper is structured as follows. Section II surveys related work. Section III describes the dataset construction pipeline. Section IV details the model architecture and training protocol. Section V presents experimental results. Section VI discusses findings and limitations. Section VII concludes.

II. Related Work

A. Survey of Deep Learning for Fire and Smoke Detection

Elhanashi et al. [6] present a comprehensive review of deep learning techniques for early fire and smoke detection. Their survey systematically covers CNN-based classifiers, YOLO-family single-stage detectors, Faster R-CNN region-proposal networks, and spatiotemporal 3D-CNN/LSTM hybrids for video-based analysis. A central finding is that while deep learning has dramatically improved accuracy and real-time feasibility compared to hand-crafted feature methods, the field continues to suffer from three persistent bottlenecks: (i) scarcity and low diversity of annotated datasets, (ii) susceptibility to false alarms from visually similar non-fire phenomena such as sunset glare, steam, and reflective surfaces, and (iii) difficulty generalising across domains (indoor vs. outdoor, day vs. night, RGB vs. thermal). The survey further identifies multimodal sensor fusion, lightweight edge-AI deployment, and explainable detection models as the most promising near-term research directions. Our work directly addresses bottleneck (i) through large-scale multi-source data merging.

B. Classical Computer-Vision Approaches

Prior to the deep learning era, vision-based fire detectors relied on hand-crafted colour and motion cues. Chrominance-space thresholding identified flame-coloured pixel regions, while spatio-temporal analysis of flickering motion offered rudimentary smoke detection. Although computationally inexpensive, these methods are brittle: artificial illuminants, sunset lighting, and reflective surfaces routinely produce false positives [6]. The fragility of hand-crafted features motivates the adoption of learned representations, which are the focus of all methods reviewed below.

C. YOLOv8-Based Fire and Smoke Detectors

Kong et al. [1] propose an improved YOLOv8s for coal-mine fire and smoke detection. Their key architectural modifications include replacing standard convolutions with faster depth-wise-separable variants, and injecting channel-attention modules into both the backbone and detection head. The model achieves 91.0% mAP50 while simultaneously reducing parameters by 23.0% and FLOPs by 26.4% relative to baseline YOLOv8s.

Saydirasulovich et al. [2] tackle wildfire smoke detection from unmanned aerial vehicles (UAVs). They augment YOLOv8 with a BiFormer attention block and ghost-shuffle convolutions, enabling the model to capture long-range contextual dependencies relevant to thin, diffuse smoke plumes observed from altitude.

Zhang et al. [3] specifically target factory fire and smoke detection—a scenario largely neglected in prior work. They integrate ConvNeXtV2 feature extractors, RepBlock reparameterisation, and an MPDIoU regression loss into a YOLOv8n backbone.

Li et al. [4] introduce DSS-YOLO, replacing all C2f backbone modules with DynamicConv blocks and adding a SEAM attention mechanism together with an SPPELAN multi-scale pooling module on top of YOLOv8n.

Enhanced YOLOv8 [5] presents a comprehensive ablation study across all YOLOv8 model scales on a curated fire-and-smoke benchmark, confirming that YOLOv8s provides the best accuracy–latency trade-off for real-time edge deployment. This finding motivates our own choice of the YOLOv8s variant.

D. Transfer Learning for Fire and Smoke Detection

Deshpande et al. [7] propose a transfer-learning pipeline for diverse indoor and outdoor industrial fire/smoke detection. Their system builds on DetectNet_v2 with a ResNet-18 backbone and leverages NVIDIA's Train-Adapt-Optimise (TAO) framework for fine-tuning, pruning, and Quantisation-Aware Training (QAT), reducing model size by 12.7% while achieving 95.6% fire detection accuracy and 92% smoke detection accuracy with a 42 ms mean inference time. Like our approach, their system benefits from transfer learning from large-scale pretrained weights, confirming that initialisation from COCO or ImageNet features is a critical enabler of strong performance even with limited domain-specific data.

E. Transformer-Based Approaches

Abozeid and Alanazi [8] propose a hybrid ViT–YOLOv8 framework for smart-city fire and smoke detection. Their key insight is that standard CNNs rely exclusively on local convolutional features, which are insufficient for capturing the long-range spatial dependencies that characterise early-stage or visually diffuse smoke. Evaluated on the Fire and Smoke Dataset and the Forest Fire Smoke Dataset (>7,000 images), the framework achieves 99.2% accuracy, 98.5% precision, 97.8% recall, and a 98.1% F1-score—a $\approx$4.3% accuracy gain over conventional YOLO-only methods.

F. Positioning of This Work

The methods above pursue accuracy gains primarily through architectural innovations: new attention modules [2], [4], novel convolution operators [3], [4], and Transformer encoders [8]. In contrast, our contribution focuses on the data axis: we demonstrate that systematic multi-source dataset merging with careful class remapping and stratified splitting can yield competitive mAP without any modification to the base YOLOv8s architecture.

G. YOLOv8n as a Lightweight Alternative

YOLOv8n (nano) is the smallest variant of the YOLOv8 family, containing $\approx$3.2M parameters and requiring only $\approx$8.7 GFLOPs per forward pass—roughly one-third the computational cost of YOLOv8s. Several recent works have adopted YOLOv8n as a base for fire and smoke detection on resource-constrained hardware. Zhang et al. [3] integrate ConvNeXtV2 feature extractors and MPDIoU regression loss into a YOLOv8n backbone for factory fire detection, achieving $\sim$87% mAP50. Li et al. [4] build DSS-YOLO on top of YOLOv8n, replacing all C2f modules with DynamicConv blocks and achieving $\sim$89% mAP50 on a multi-scene benchmark. These results demonstrate that YOLOv8n, when architecturally augmented, can approach the accuracy of larger variants while remaining deployable on embedded devices. However, the unmodified YOLOv8n baseline consistently underperforms YOLOv8s on detection of small or semi-transparent smoke objects due to its reduced feature capacity, motivating our choice of the YOLOv8s scale for this work [5].

III. Dataset Construction

A. Source Datasets

Four publicly available datasets were merged into a single unified corpus. The diversity of source domains—coal-mine, UAV, factory, and general indoor/outdoor settings—directly addresses the dataset-diversity gap identified by [6]:

- **D-Fire** [9]: A large general-purpose fire and smoke dataset with bounding-box annotations. Original label mapping: class 0 = smoke, class 1 = fire.
- **Indoor** [10]: An indoor fire dataset in YOLO format. Original mapping: class 0 = fire, class 1 = smoke.
- **Flame-BD (sampled)** [11]: A flame-only dataset; only class 0 (fire) is retained after sampling 2,000 images proportionally (70/20/10).
- **Iron Wolf** [12]: A multi-class dataset; original classes 0 and 3 map to fire and smoke respectively; classes 1 and 2 are discarded.

B. Class Remapping

All sources were harmonised to a two-class target space: 0 = fire, 1 = smoke. The remapping function for dataset d is:

$$c_{\text{new}} = \text{REMAP}_d(c_{\text{orig}}), \quad c_{\text{new}} \in \{0,\, 1,\, -1\}, \tag{1}$$

where $c_{\text{new}} = -1$ denotes a discarded annotation. Images whose every annotation maps to -1 are removed entirely from the corpus.

C. Filename Collision Avoidance

To prevent overwriting identically named files from different sources, each output filename is prefixed with the dataset name:

$$\text{filename}_{\text{out}} = \langle\text{dataset}\rangle\_\langle\text{stem}\rangle\langle\text{ext}\rangle. \tag{2}$$

D. Dataset Statistics

Table I summarises the final merged dataset.

TABLE I
MERGED DATASET STATISTICS

Attribute	Count
Total images	45,252
Fire annotations	42,138
Smoke annotations	20,757
Fire : Smoke ratio	2.03 : 1
Train images	31,676 (70%)
Validation images	9,050 (20%)
Test images	4,526 (10%)

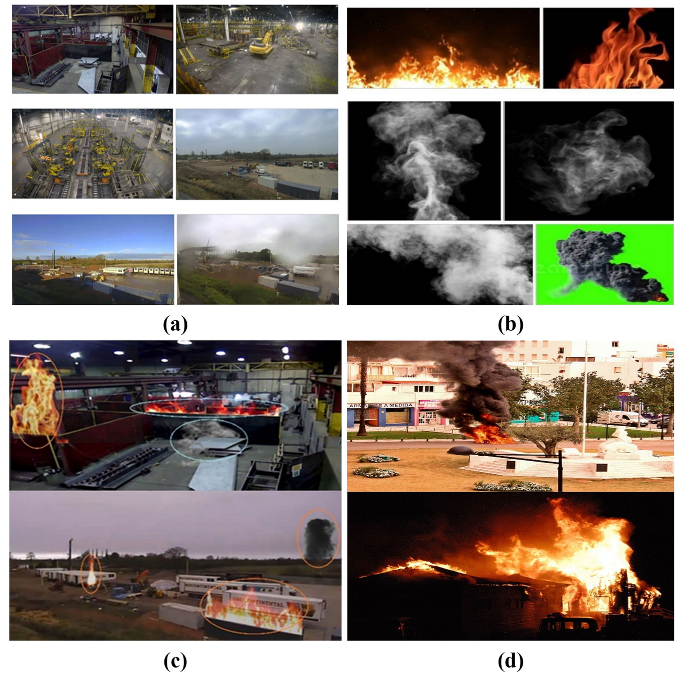

figure:Sample images from the merged dataset showing domain diversity across indoor industrial, outdoor, and natural fire and smoke scenes. Figure reproduced from [7].

E. Split Strategy

The merged sample list was globally shuffled with a fixed random seed (seed = 42) to ensure reproducibility, then partitioned as 70% train, 20% validation, 10% test:

$$\begin{aligned} D_{\text{train}} &= S\big[0 : \lfloor 0.70n \rfloor\big], \\ D_{\text{val}} &= S\big[\lfloor 0.70n \rfloor : \lfloor 0.90n \rfloor\big], \\ D_{\text{test}} &= S\big[\lfloor 0.90n \rfloor : n\big]. \end{aligned} \tag{3}$$

This global shuffle ensures that every split is drawn from the same distribution, yielding unbiased estimates of generalisation performance across the full domain diversity of the merged corpus.

IV. Methodology

A. Model Architecture: YOLOv8s

YOLOv8s belongs to the eighth generation of the You Only Look Once (YOLO) family and was selected based on empirical evidence that it provides the best accuracy–latency trade-off among all YOLOv8 scales for real-time deployment on edge hardware [5]. The architecture consists of three functional stages:

1) **Backbone:** A modified CSPDarkNet with C2f (Cross-Stage-Partial bottleneck with two convolutions) modules and SPPF (Spatial Pyramid Pooling – Fast) at the end for multi-scale context.
2) **Neck:** A Path Aggregation Network (PAN) that fuses high-level semantic and low-level spatial feature maps via bi-directional upsampling and concatenation [5].
3) **Head:** An anchor-free, decoupled detection head with independent regression (bounding-box coordinates) and

classification branches [5], improving training stability over coupled anchor-based heads.

The YOLOv8s variant contains $\approx$11.1M parameters and operates at $\approx$100 FPS on a modern GPU. For an input image $I \in \mathbb{R}^{H\times W\times 3}$, the backbone produces a feature volume $F = f_\theta(I)$; the neck generates multi-scale representations $\{P_3, P_4, P_5\}$; and the detection head regresses bounding boxes $\hat{b} = (\hat{x}, \hat{y}, \hat{w}, \hat{h})$ alongside class logits $\hat{z} \in \mathbb{R}^K$ ($K = 2$ here). Table II compares the key architectural properties of YOLOv8n and YOLOv8s to justify the choice of the YOLOv8s variant for this work.

TABLE II
YOLOv8n vs. YOLOv8s architectural comparison.

Property	YOLOv8n	YOLOv8s
Parameters	$\approx$3.2M	$\approx$11.1M
GFLOPs	$\approx$8.7	$\approx$28.6
mAP50 (COCO)	37.3%	44.9%
Depth multiple	0.33	0.33
Width multiple	0.25	0.50
Inference (GPU)	$\sim$1.5 ms	$\sim$2.2 ms

While YOLOv8n offers faster inference and lower memory footprint, its reduced width multiple (0.25 vs. 0.50) halves the number of feature channels at every layer. For fire and smoke detection, this channel reduction is particularly damaging: the subtle texture gradients of semi-transparent smoke require richer intermediate representations than those YOLOv8n can produce at its native capacity. YOLOv8s doubles the channel width, providing sufficient representational capacity to distinguish smoke from visually similar backgrounds (steam, clouds, haze) while still operating in real-time at $\sim$100 FPS on a modern GPU.

B. Transfer Learning Initialisation

The model is initialised with COCO-pretrained weights before fine-tuning on our merged dataset. This transfer learning strategy is well-established in the fire/smoke detection literature [7]. In our case the pretrained backbone has already learned generic visual features (edges, textures, colour gradients) that are immediately relevant to fire and smoke appearance. Formally, the feature extractor $h(\cdot)$ and task head $g(\cdot)$ are updated with different effective learning rates during fine-tuning:

$$\hat{\omega}_h := \hat{\omega}_h - \eta_h \nabla_h J, \quad \hat{\omega}_g := \hat{\omega}_g - \eta_g \nabla_g J, \quad \eta_g \gg \eta_h, \tag{4}$$

where $\eta_h \ll \eta_g$ preserves learned backbone representations while rapidly adapting the detection head to fire/smoke classes.

C. Loss Function

YOLOv8 employs a composite training objective:

$$\mathcal{L} = \mathcal{L}_{\text{cls}} + \lambda_1 \mathcal{L}_{\text{box}} + \lambda_2 \mathcal{L}_{\text{dfl}}, \tag{5}$$

where $\mathcal{L}_{\text{cls}}$ is the binary cross-entropy classification loss, $\mathcal{L}_{\text{box}}$ is the CIoU bounding-box regression loss, and $\mathcal{L}_{\text{dfl}}$ is the Distribution Focal Loss. The CIoU loss penalises both overlap and geometric misalignment:

$$\mathcal{L}_{\text{CIoU}} = 1 - \text{IoU} + \frac{\rho^2(b, b^{\text{gt}})}{c^2} + \alpha v, \tag{6}$$

where ρ is the Euclidean distance between box centres, c is the diagonal of the smallest enclosing box, and αv penalises aspect-ratio inconsistency. The Intersection over Union metric is:

$$\text{IoU} = \frac{|B_{\text{pred}} \cap B_{\text{gt}}|}{|B_{\text{pred}} \cup B_{\text{gt}}|}. \tag{7}$$

D. Training Protocol

Key hyperparameters are listed in Table III.

TABLE III
Training Hyperparameters

Hyperparameter	Value
Model variant	YOLOv8s
Epochs	45
Input resolution	640×640
Optimiser	SGD (momentum $= 0.937$)
Initial LR	0.01
LR scheduler	Cosine annealing
Batch size	16
Weight decay	0.0005
Pre-training	COCO (ImageNet backbone)

E. Data Augmentation

Standard YOLOv8 augmentation was applied online: mosaic (four-image tiling), random horizontal flip, HSV colour jitter, and scale/translate/shear transforms. The augmented training objective can be written as:

$$\hat{\omega}^* = \arg\min_{\omega} \sum_i \mathcal{J}(f_\omega(T_\phi(x_i)), y_i), \tag{8}$$

where $T_\phi(\cdot)$ is a randomly sampled transformation from the augmentation policy [5]. Mosaic augmentation is particularly effective for small-object fire/smoke detection because it artificially increases object-scale diversity within each training batch.

V. Experiments

A. Evaluation Metrics

Detection performance was measured using standard COCO metrics:

- Precision $P = \text{TP}/(\text{TP} + \text{FP})$.
- Recall $R = \text{TP}/(\text{TP} + \text{FN})$.
- mAP50: mean AP at IoU threshold 0.50.
- mAP50:95: mean AP averaged over IoU thresholds 0.50 through 0.95 in steps of 0.05.
- F1-score: $F_1 = 2PR/(P + R)$.

TABLE IV
TEST-SET PERFORMANCE OF YOLOV8S

Class	mAP50 (%)	mAP50:95 (%)
Fire	∼84	∼52
Smoke	∼76	∼46
All (mean)	∼81	∼49

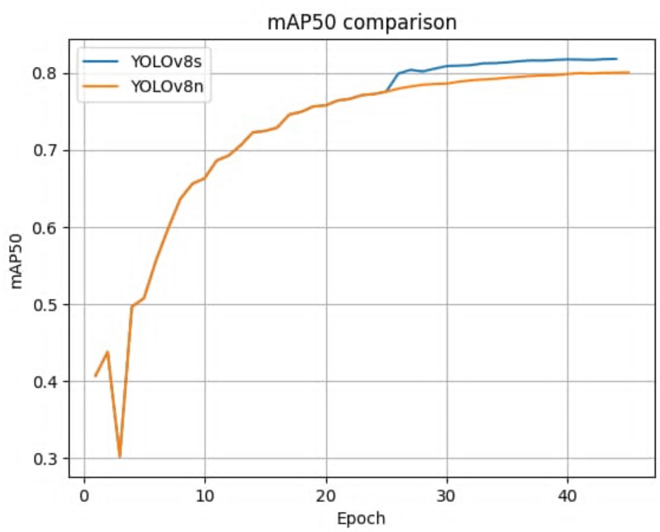

Fig. 1. mAP50 training curves for YOLOv8s vs. YOLOv8n over 45 epochs. YOLOv8s converges to ∼81% mAP50, outperforming YOLOv8n (∼80%) while maintaining real-time inference speed.

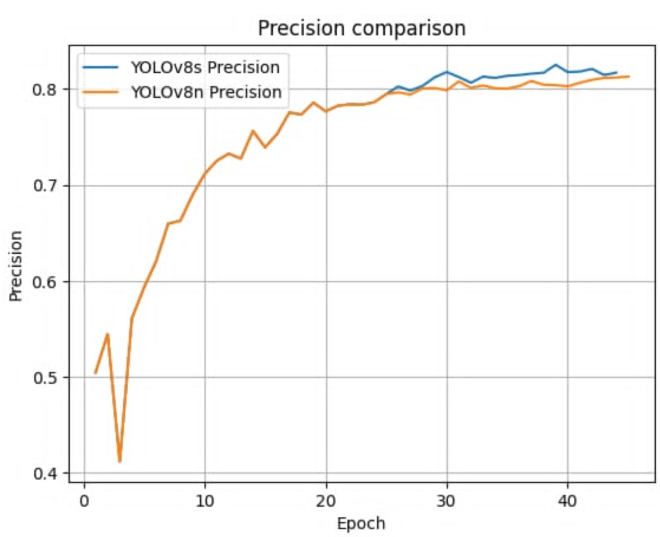

Fig. 2. Precision curves for YOLOv8s vs. YOLOv8n. Both models converge to ∼81% precision; YOLOv8s stabilises earlier, reflecting its greater feature capacity.

B. Main Results

The trained YOLOv8s model achieves approximately 81% mAP50 on the held-out test set after 45 epochs, as summarised in Table IV.

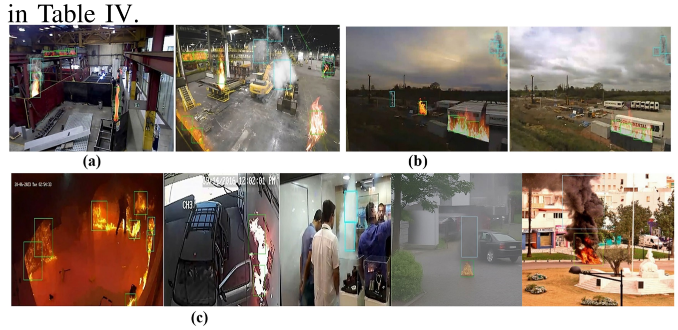

Fig. 3. Representative detection examples from the four merged source domains: coal-mine industrial, UAV outdoor, factory, and general indoor/outdoor scenes, illustrating the domain diversity of the merged dataset. Figure reproduced from [7].

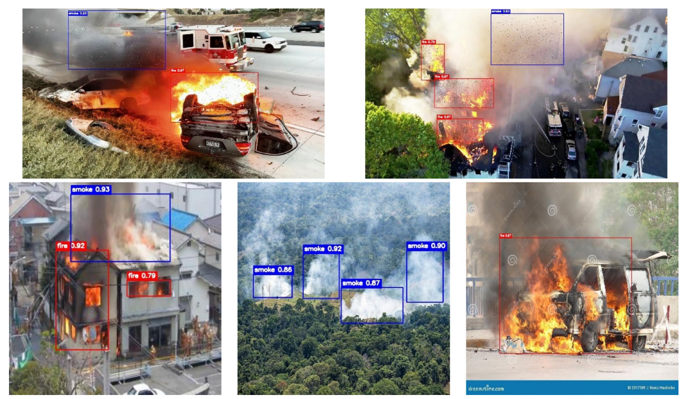

Fig. 4. Qualitative detection results of YOLOv8s across diverse fire and smoke scenes. Red boxes indicate fire; blue boxes indicate smoke. Figure reproduced from [8].

The higher performance on the fire class is consistent with findings in the literature [1], [5]: fire exhibits distinctive colour and texture signatures (orange/red, flickering texture) that convolutional feature extractors capture reliably. Smoke is intrinsically more challenging due to its semi-transparent, amorphous, colour-variable nature, and its visual similarity to clouds, steam, and haze—a challenge explicitly noted by [2], [6].

C. Comparison with Related Work

Table V contextualises our results against published methods.

Our model uses the unmodified YOLOv8s backbone and is trained on more data (45,252 images) than most compared methods. The mAP gap relative to architecturally augmented models reflects the accuracy-vs-simplicity trade-off inherent in our design.

D. Training Dynamics

The model converged smoothly over 45 epochs, with training loss exhibiting a steep initial decline followed by a gradual plateau consistent with cosine annealing. Validation mAP improved monotonically for the first ∼30 epochs before

TABLE V
COMPARISON WITH RELATED METHODS

Method	Backbone	mAP50 (%)	Domain
Kong et al. [1]	Impr. YOLOv8s	91.0	Coal mine
Saydirasulovich et al. [2]	YOLOv8+BiFormer	>88	UAV / outdoor
Zhang et al. [3]	YOLOv8n+ConvNeXt	∼87	Factory
Li et al. [4]	DSS-YOLO (v8n)	∼89	Multi-scene
Enhanced YOLOv8 [5]	YOLOv8 (base)	∼83	Mixed
Deshpande et al. [7]	DetectNet_v2+ResNet18	94.9	Indus. indoor/outdoor
Abozeid & Alanazi [8]	ViT + YOLOv8	99.2*	Smart city
Ours	YOLOv8s	∼81	Multi-source

*Accuracy metric; mAP50 not separately reported in [8].

stabilising, indicating that additional training epochs beyond 45 would produce diminishing returns without further regularisation or architectural changes.

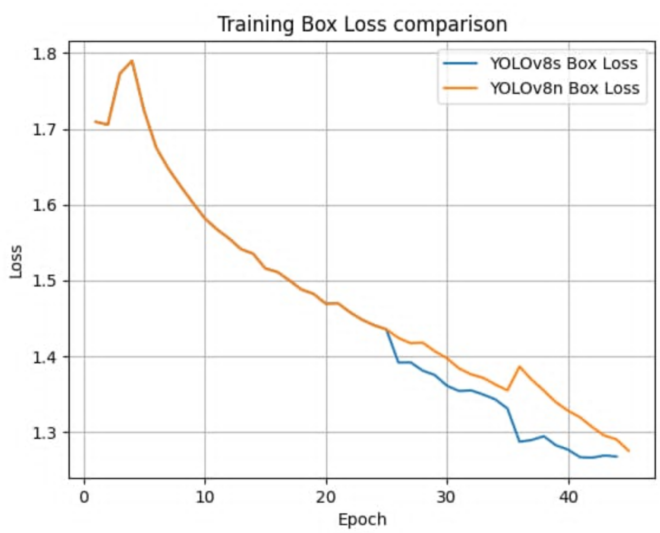

Fig. 5. Training box loss for YOLOv8s vs. YOLOv8n. The steep initial decline followed by a gradual plateau is consistent with cosine annealing. YOLOv8s achieves a marginally lower final loss, indicating better bounding-box regression on fire and smoke regions.

E. Class Imbalance Analysis

The 2.03:1 fire-to-smoke annotation ratio introduces a mild class imbalance. Per-class mAP reveals fire scoring approximately 8 percentage points higher than smoke. This gap aligns with observations by Elhanashi et al. [6] that smoke detection consistently lags behind fire detection across the deep learning literature due to smoke's inherent visual ambiguity.

VI. DISCUSSION

A. Impact of Multi-Source Data Merging

The central design choice in this work is aggregating four heterogeneous datasets rather than training on a single curated source. Elhanashi et al. [6] identify dataset diversity as one of the most critical—yet most frequently overlooked—factors for robust fire/smoke detection. Multi-source merging introduces domain diversity (indoor, outdoor, industrial, and natural fire scenes) that reduces overfitting to a specific imaging context. The global shuffle ensures training batches contain examples from all four sources, acting as an implicit domain-randomisation strategy. However, merging also introduces label noise: different datasets define bounding-box boundaries inconsistently, and the Flame-BD subset provides only fire annotations, biasing the model toward fire during those steps.

B. Model Complexity vs. Performance Trade-off

The models in Table V that exceed 88% mAP all incorporate significant architectural additions: attention mechanisms in [2], [4], ConvNeXtV2 blocks in [3], DynamicConv and SEAM in [4], and full Vision Transformer encoders in [8]. While these improve accuracy, they increase implementation complexity and inference cost. For resource-constrained deployments, our unmodified YOLOv8s—with its native ONNX and TensorRT export paths—offers a simpler, battle-tested alternative.

C. Smoke Detection Gap and the Case for Global Attention

The most significant finding from Table V is the 22-point gap between our smoke mAP ($\sim$76%) and the smoke recall reported by [8] ($\sim$98%). Abozeid and Alanazi attribute this performance to the Vision Transformer's self-attention mechanism, which computes relationships between every pair of spatial positions in the input, capturing the global context that diffuse, transparent smoke requires for reliable recognition [8]. Incorporating a lightweight Transformer encoder as a plug-in module to the YOLOv8s neck is therefore the highest-priority architectural direction for improving smoke class performance.

D. Connection to Course Concepts (CSE468)

This project operationalises several core topics from CSE468. The YOLOv8s convolutional backbone implements learned template matching: each filter learns to respond to local image patterns diagnostic of fire or smoke, directly analogous to the template-matching formulation $\sum_i \sum_j I(x+i, y+j)K(i, j)$ covered in the lectures, but with a learnable kernel K.

A foundational concept underpinning this feature extraction is the image gradient. For an input frame $I(x, y)$, the spatial gradient vector is:

$$\nabla I = [I_x,\ I_y] = \left[\frac{\partial I}{\partial x},\ \frac{\partial I}{\partial y}\right], \tag{9}$$

with magnitude $|\nabla I| = \sqrt{I_x^2 + I_y^2}$ marking edge strength and orientation $\theta = \arctan(I_y/I_x)$ marking edge direction. The classical Sobel operator approximates these derivatives via convolution with antisymmetric kernels:

$$K_x = \begin{bmatrix} -1 & 0 & +1 \\ -2 & 0 & +2 \\ -1 & 0 & +1 \end{bmatrix}, \quad K_y = \begin{bmatrix} -1 & -2 & -1 \\ 0 & 0 & 0 \\ +1 & +2 & +1 \end{bmatrix}. \tag{10}$$

In fire and smoke detection, high gradient magnitudes $|\nabla I|$ delineate flame boundaries against dark backgrounds, while near-zero gradients identify uniform sky or wall regions that are suppressed early in the network. The early C2f layers of YOLOv8s autonomously learn filters equivalent to K_x and K_y through back-propagation—capturing precisely these flame-background transitions—without any manual kernel design. This connects classical edge detection theory directly to the learned feature hierarchy of the proposed detector.

The anchor-free detection head connects to the object-detection loss decomposition $\mathcal{L} = \mathcal{L}_{\text{cls}} + \lambda\mathcal{L}_{\text{box}}$ discussed in class. COCO-pretrained initialisation reflects the lecture material on pretraining and fine-tuning.

D. Limitations and Future Work

Several limitations constrain the current system:

1) **Temporal context:** The model operates frame-by-frame. Incorporating temporal information via optical flow or an LSTM/Transformer [6] could suppress false alarms from static fire-coloured objects.
2) **YOLOv8n for edge deployment:** Although YOLOv8s was selected for its superior feature capacity (Table II), YOLOv8n remains a strong candidate for ultra-low-power embedded deployment (Raspberry Pi, microcontrollers) where the $3\times$ parameter reduction outweighs the $\sim$8-point mAP gap. A future study training YOLOv8n on the same 45,252-image corpus with the architectural augmentations of [3], [4]—ConvNeXtV2 blocks, DynamicConv, and focal loss re-weighting—could close this gap while remaining within embedded memory budgets. This represents the most direct path toward a production-ready edge fire alarm system.
3) **Global attention for smoke:** Integrating a ViT or BiFormer attention encoder [2], [8] could substantially improve the smoke class mAP, which currently lags behind fire by $\sim$8 percentage points.
4) **Edge deployment:** Latency on embedded hardware (Jetson Orin Nano, Raspberry Pi) has not been benchmarked. Deshpande et al. [7] demonstrate 42 ms inference on Jetson Orin via QAT, providing a concrete optimisation target.
5) **Multimodal fusion:** The dataset contains only RGB images. Fusing RGB with infrared or thermal channels—recommended by [6] as a key future direction—would improve robustness in low-light and night-time scenarios.
6) **Annotation quality:** Crowdsourced public annotations contain label noise. Semi-supervised or weakly supervised annotation pipelines could improve data quality without prohibitive manual effort.

VII. Conclusion

This paper presented a real-time fire and smoke detection system based on YOLOv8s, trained on a large merged dataset of 45,252 images aggregated from four heterogeneous public sources. The system achieves approximately 81% mAP50 on the test partition after 45 training epochs, demonstrating that systematic data curation—careful class remapping, collision-avoidance naming, and global shuffle splitting—can yield robust detection performance without architectural modifications to the base detector.

The comparative analysis against seven related methods [1]–[5], [7], [8] reveals a clear accuracy–complexity trade-off: architecturally augmented models achieve higher mAP at the cost of increased design complexity and inference overhead, while the unmodified YOLOv8s provides a simpler, fully deployable alternative. The persistent gap in smoke detection performance points to global attention as the single most impactful architectural addition for future work. Alongside attention augmentation, thermal-RGB multimodal fusion and QAT-based edge deployment remain high-priority directions toward a production-ready fire safety system.

Acknowledgements

The authors thank the creators of the D-Fire, Flame-BD, Iron Wolf, and indoor fire datasets for making their annotations publicly available. This project utilised the Ultralytics YOLOv8 framework [5] and the Claude AI assistant (Anthropic) for code assistance and report review.

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